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A noise-resilient distributed recurrent neural network for multi-agent consensus control and acoustic source localization^{*}

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ABSTRACT

In multi-agent systems (MASs), consensus control and acoustic source localization are fundamental yet noise-sensitive tasks. While recurrent neural networks (RNNs) have shown strong potential in these domains due to their dynamic modeling capabilities, their control accuracy degrades notably under noise interference, and RNN-based control for disturbed multi-agent scenarios has received limited attention. To address this challenge, we design a noise-resilient distributed RNN (NDRNN) within the multi-agent consensus framework and develop a corresponding consensus control protocol, termed NDRNN-CP. The NDRNN incorporates a time-delay mechanism to adaptively learn noise variation patterns and employs an optimized activation function for accelerated convergence. Rigorous theoretical analyses prove that NDRNN-CP ensures global stability, robustness against periodic and stochastic disturbances, and predefined-time convergence. Furthermore, building on the NDRNN design principle, we propose NDRNN-S, a distributed acoustic source localization solver capable of maintaining high accuracy in noisy multi-agent environments. Extensive simulations on multi-agent consensus and distributed acoustic source localization demonstrate that, compared with conventional DRNN-based methods, the proposed NDRNN-CP and NDRNN-S achieve faster convergence and significantly lower steady-state errors under various noise signal conditions, confirming the effectiveness and broad applicability of the proposed approach.

1. Introduction

MASs (Chen et al., 2026; Wu et al., 2025) are increasingly deployed in advanced applications such as multi-aircraft (Liang et al., 2026), mobile robots (Li et al., 2024; Yang et al., 2025), surface vehicles (Wang et al., 2025a), and distributed sensing networks (He et al., 2023), where coordinated operation is essential for ensuring stability and efficiency (Cao et al., 2025; Wang et al., 2025b; Zhang et al., 2025). Two representative challenges in this domain are consensus control (Lu et al., 2024), which aims to design control laws that drive all agents to a common state, and acoustic source localization (Zhang et al., 2024b), which supports tasks ranging from robot audition to speech signal enhancement. RNNs (Yan et al., 2024), derived from continuous Hopfield networks, have emerged as a powerful paradigm for such multi-agent coordination tasks due to their inherent dynamic modeling capabilities and distributed architecture. However, a critical barrier persists across these approaches: in realistic scenarios, both classical and RNN-based methods suffer a severe loss of accuracy and stability in the presence of

noise, external disturbances, or reverberation. For consensus control, a variety of controllers have been developed (Lee et al., 2025; Yuan et al., 2025), their performance degrades under non-ideal conditions. Similarly, for acoustic source localization, techniques like time-delay estimation (Kocur et al., 2024) and Kalman filtering offer the optimal Bayesian estimation (Tian et al., 2023) are vulnerable to the nonlinearities inherent in practical scenarios, such as those involving time-difference-of-arrival (TDOA) measurements (Manamperi et al., 2022; Wang et al., 2024).

To enhance noise resilience, particularly within the promising RNN framework, research has predominantly followed three strategies. The first employs specialized activation functions (AFs) with intrinsic noise-suppressing properties (Cao et al., 2024; Dai et al., 2021), though these can introduce discontinuities and may yield suboptimal accuracy. The second strategy incorporates integral terms (Jin et al., 2015; Liao et al., 2024), which are effective against constant or low-amplitude bounded noise but are inadequate for more complex, time-varying disturbances. A third approach integrates adaptive components to learn the charac-

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teristics of cyclically varying noise (Guo et al., 2019; Li et al., 2023; Liao et al., 2022; Luo et al., 2021; Zhang et al., 2019), yet these remain limited in general noise scenarios. When applied to MASs, these solutions often present a difficult trade-off: most methods rely on AFs with only modest robustness (Cao et al., 2024; Li et al., 2025; Luo et al., 2023), while others, like fuzzy controllers (Dai et al., 2022; Jia et al., 2023), improve noise suppression at the cost of significantly increased computational complexity, thereby impairing real-time performance. These limitations highlight a clear gap: there is a need for a distributed RNN framework that combines strong noise immunity, enabling reliable consensus control and acoustic source localization in disturbed multi-agent environments.

To address these challenges, this work develops a NDRNN framework tailored for multi-agent systems operating in disturbed environments. This framework is first realized as a consensus control protocol, termed NDRNN-CP, which introduces two principal innovations: a time-delay mechanism that functions as an internal model to reject periodic disturbances of a known period, and a novel AF that guarantees system convergence within a predefined time. The contributions are supported by rigorous theoretical analysis that establishes the global stability, robustness, and predefined-time convergence of the NDRNN-CP. To demonstrate the practical utility and versatility of this architecture, we further apply its core principles to design an NDRNN-based solver (NDRNN-S) for the challenging task of distributed acoustic source localization. This solver enables agents to collaboratively track a source in noisy and reverberant conditions using only locally exchanged information. Extensive simulations on multi-agent consensus and 3-D acoustic source localization demonstrate that NDRNN-CP and NDRNN-S consistently outperformed existing RNN-based approaches, exhibiting predefined-time convergence with enhanced noise tolerance and robustness across complex multi-agent coordination and distributed sensing task. This paper builds upon our previous conference paper (Li et al., 2025) and substantially extends the conference version by establishing a unified NDRNN framework, introducing a new application to distributed acoustic source localization, strengthening the theoretical analysis, and providing more extensive simulations and experiments.

The main contributions of this paper are as follows.

- We propose a NDRNN framework, designed from the consensus problem perspective, which incorporates a time-delay component to adaptively learn noise variation patterns and integrate noise signal processing into the control architecture to achieve enhanced disturbance suppression.
- Based on the NDRNN framework, we develop two methods: a consensus control protocol (NDRNN-CP) that enables all agents to reach consensus within a predefined time under diverse noise conditions, and an NDRNN-based solver (NDRNN-S) for distributed acoustic source localization that achieves high accuracy and robustness in noisy and reverberant environments.
- We design a new AF specifically to ensure predefined-time convergence, and, using the consensus problem as the theoretical basis, rigorously prove the global stability, robustness, and predefined-time convergence of the proposed NDRNN framework.
- Extensive simulations and experiments demonstrate that, compared with mainstream DRNN-based approaches, NDRNN-CP achieves more robust predefined-time consensus in large-scale networks, while the NDRNN-S delivers superior localization accuracy in both noise-free and noisy conditions.

The remainder of this paper is structured as follows. Section 2 discusses the leader-follower consensus problem, providing the foundational context for our work. Section 3 introduces the design of common control protocols based on DRNN, with a particular focus on our proposed NDRNN-CP. Section 4 presents the theoretical analysis, including stability, robustness, and predefined-time convergence proofs. Section 5 presents the experimental evaluation, where we compare NDRNN-CP with other DRNN-based control protocols, highlighting its superior per-

formance in terms of noise resistance and convergence. Section 6 extends the NDRNN framework to distributed acoustic source localization. Finally, Section 7 concludes the paper.

2. Consensus control problem description

This section presents the leader-follower consensus problem in MASs (Lewis et al., 2013; Amini et al., 2018). In this problem, a group of agents cooperates to reach consensus, with certain agents designated as leaders and the remainder acting as followers. The goal of the consensus problem is to drive all agents' states to coincide, ensuring that followers match the states of the leaders. Importantly, followers may not have direct communication with leaders; instead, information exchange follows a predefined inter-agent topology. Table 1 summarizes the main symbols used throughout the paper, together with their units, types, and brief descriptions. Common basics (e.g., t , N , i , sets such as $\mathbb{R}$, and graph-related matrices like A , D , and L) are omitted.

Consider a group of N agents, where the state of agent i at time t is denoted as $\mathbf{x}_i(t)$, for $i = 1, 2, \dots, N$. The state of the designated leader is given by $\mathbf{x}_L(t)$, and all other agents are treated as followers. The evolution of the system is governed by the following equation:

$$\dot{\mathbf{x}}_i(t) = \mathbf{u}_i(t) + \mathbf{d}_i(t), \quad i = 1, 2, \dots, N, \quad (1)$$

where $\mathbf{u}_i(t)$ denotes the control signal applied to agent i , and $\mathbf{d}_i(t)$ captures the external disturbance or noise affecting its dynamics. The leader follows the dynamics $\dot{\mathbf{x}}_L(t) = \mathbf{u}_L(t)$, with $\mathbf{u}_L(t)$ given in advance.

The aim is to construct $\mathbf{u}_i(t)$ for each follower so that their states $\mathbf{x}_i(t)$ asymptotically coincide with the leader's state $\mathbf{x}_L(t)$, thereby realizing leader-follower consensus, i.e.,

$$\lim_{t \rightarrow \infty} \|\mathbf{x}_i(t) - \mathbf{x}_L(t)\|_2 = 0, \quad i = 1, 2, \dots, N. \quad (2)$$

By adaptively adjusting the control actions of the followers according to the given communication topology, robust consensus can be maintained even in the presence of noise signal disturbances.

Since both periodic noise and random noise are commonly used in control applications, this work considers several representative types: sine wave, square wave, sawtooth wave, constant, zero-mean random noise, and composite periodic noise signals. The waveforms of these noise functions are illustrated in Fig. 1.

3. DRNN-based control protocols

In this section, we first design the proposed NDRNN from the perspective of the consensus problem. Based on this framework, we develop the NDRNN-CP, which guarantees that all followers reach consensus with the leader within a predefined time under various noise conditions. For comparative analysis, we also present two baseline protocols, namely the ODRNN-CP and the IDRNN-CP.

In the NDRNN-CP design, the communication topology plays a critical role. Since not all followers have direct access to the leader and the leader's trajectory is predefined, the control law focuses on the followers. To characterize the state deviations that drive the control protocol, two error functions are defined for each follower i .

- Inter-follower error function:

$$\mathbf{e}_{f_i}(t) = \sum_{j \in \mathcal{N}_i} (\mathbf{x}_i(t) - \mathbf{x}_j(t)), \quad (3)$$

where $\mathcal{N}_i$ denotes the set of neighboring followers that can communicate directly with follower i .

- Leader-follower error function:

$$\mathbf{e}_{l_i}(t) = (\mathbf{x}_i(t) - \mathbf{x}_L(t))\delta(i), \quad (4)$$

where $\delta(i)$ denotes an indicator function, with $\delta(i) = 1$ if follower i can directly communicate with the leader, and $\delta(i) = 0$ otherwise.

Table 1
Nomenclature.

Category	Symbol	Description
Controller-specific terms	$e_{f_i}(t)$	Neighbor consensus error $\sum_{j \in \mathcal{N}_i} (\mathbf{x}_i(t) - \mathbf{x}_j(t))$.
	$e_{l_i}(t)$	Leader-follower error $(\mathbf{x}_i(t) - \mathbf{x}_L(t)) \delta(i)$.
	$\delta(i)$	Indicator: 1 if agent i can directly communicate with the leader.
	$g_i(t)$	Nonlinear term $\alpha_1 \phi(e_{f_i}(t)) + \alpha_2 \phi(e_{l_i}(t))$.
	$\gamma_i(t)$	Internal periodic term: $\gamma_i(t) = \gamma_i(t-T) + \alpha_3 \phi(e_{f_i}(t) + e_{l_i}(t))$ for $t \geq T$.
	$\alpha_1, \alpha_2, \alpha_3$	NDRNN controller gains.
	μ_1, μ_2	Integral gains in IDRNN-CP.
Activation functions	$\phi(\cdot)$	Odd, strictly increasing predefined-time activation function.
	$\beta_1, \beta_2, k, \tau$	Parameters of $\phi(\cdot)$ (shape and scaling), with $\tau \in (0, 1)$.
	$\Phi(\cdot)$	Elementwise extension of $\phi(\cdot)$ to vectors.
Graph augmentation and stability	B	Diagonal leader-access matrix with $b_i \in \{0, 1\}$.
	H (graph)	Augmented graph matrix $H = L + B$.
	$V(t)$	Lyapunov function $V(t) = \frac{1}{2} e_L^T(t) H e_L(t)$ (stacked leader-follower error e_L).
	$\lambda_{\min}(H), \lambda_{\max}(H)$	Minimum/maximum eigenvalues of H (bounds in analysis).
	ϵ	Inequality tuning parameter (Young/Cauchy-Schwarz).
TDOA/DASL subsystem	v	Speed of sound used in TDOA.
	$\Delta T_{ij}(t)$	TDOA between microphones i and j .
	$A_i, \mathbf{b}_i(t)$	Local linear system $A_i, \mathbf{S}(t) = \mathbf{b}_i(t)$.
	H (DASL)	Global coefficient matrix assembled from all A_i .
	$\mathbf{G}(t)$	Global observation vector assembled from $\mathbf{b}_i(t)$.
	$\mathbf{E}_m(t)$	Residual $H \mathbf{S}(t) - \mathbf{G}(t)$.
	$\gamma_m(t)$	Periodic delay term in DASL (period T).

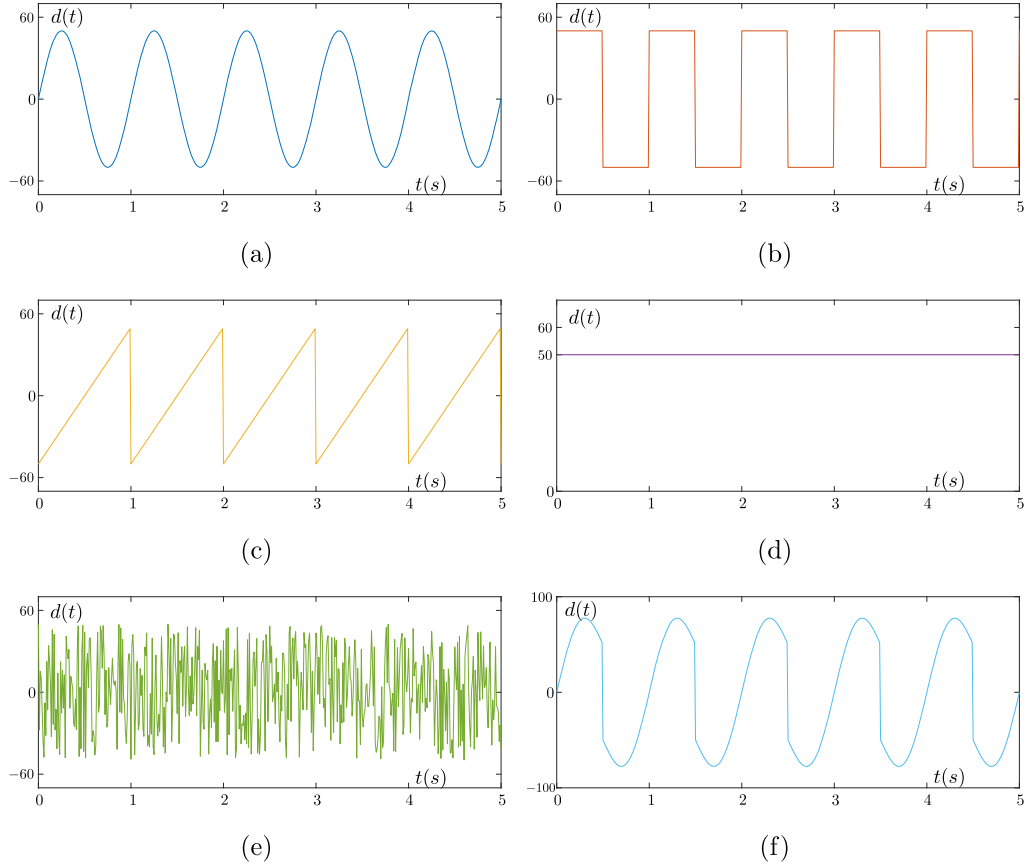


Fig. 1. Different types of noise signals. (a) Sine wave noise. (b) Square wave noise. (c) Sawtooth wave noise. (d) Constant noise. (e) Zero-mean random noise. (f) Composite period noise.

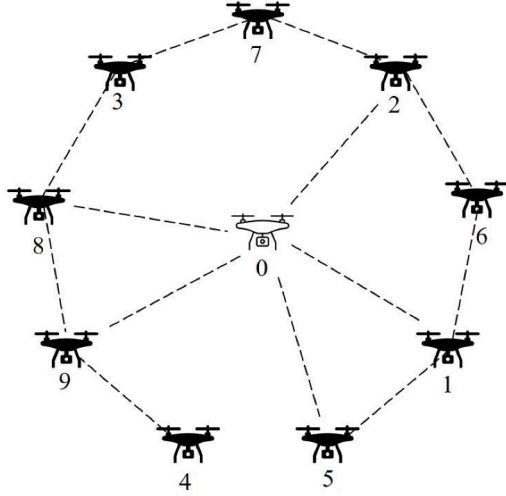


Fig. 2. Multi-agent communication topology diagram, with the white node representing the leader agent and black nodes indicating follower agents.

In accordance with established DRNN design principles (Li et al., 2023; Li, 2018; Luo et al., 2023; Petković & Stanimirović, 2024; Qiao et al., 2018; Xiao et al., 2022; Zhang & Ge, 2005), the error dynamics are driven along the negative gradient direction, with an additional time-delay term to learn and compensate for periodic noise with a known period T . The resulting error-form NDRNN is given by:

$$\dot{e}_f(t) + \dot{e}_l(t) = -\alpha_1 \phi_1(e_{f_l}(t)) - \alpha_2 \phi_2(e_{l_l}(t)) - \gamma_i(t), \quad (5)$$

where α_1 , α_2 , and α_3 are positive hyperparameters, and ϕ_1 , ϕ_2 , and ϕ_3 are nonlinear AFs applied element-wise to the vectors, and

$$\gamma_i(t) = \begin{cases} 0, & \text{if } t < T, \\ \gamma_i(t - T) + \alpha_3 \phi_3(e_{f_l}(t) + e_{l_l}(t)), & \text{otherwise.} \end{cases} \quad (6)$$

It is worth noting that in multi-agent systems subject to periodic environmental disturbances, the parameter T is typically a known constant determined by system properties or frequency analysis. During the initial phase where $t < T$, the term $\gamma_i(t)$ takes a value of zero. This condition implies that when $t < T$, the model operates as a DRNN without introducing the additional noise-suppression term.

In practical engineering scenarios such as rotating machinery or repetitive robotic tasks, the period T represents a physical property that is typically known a priori or can be estimated via frequency analysis. Regarding the initialization phase where $t < T$, the delay term $\gamma_i(t)$ remains inactive by taking a zero value. Consequently, the system temporarily functions as a standard DRNN. The stability during this interval is guaranteed by the inherent properties of the baseline DRNN established in Theorem 1, thereby ensuring bounded behavior before the active noise suppression mechanism fully engages when $t \geq T$.

These AFs play an important role in shaping the convergence behavior of the dynamic network, influencing both the rate and the effectiveness of achieving consensus (Yang & Li, 2025; Yu et al., 2024; Zhang et al., 2024a). In most cases, ϕ_1 , ϕ_2 , and ϕ_3 share the same functional form. To guarantee convergence within a predefined time, this study proposes the following AF.

$$\phi(x) = \frac{\pi}{k\sqrt{\beta_1\beta_2}}(\beta_1|x|^{1-\tau} + \beta_2|x|^{1+\tau}) \cdot \text{sign}(x), \quad (7)$$

where β_1 , β_2 and k are positive hyperparameters, $\tau \in (0, 1)$, and $\text{sign}(\cdot)$ is the sign function. The specific combination of $\beta_1|x|^{1-\tau} + \beta_2|x|^{1+\tau}$ naturally gives the terms π and $\sqrt{\beta_1\beta_2}$ in the following upper bound derivation. These two terms can cancel out the same terms in $\pi/(k\sqrt{\beta_1\beta_2})$, and thus realize predefined-time convergence. This is the difference between this AF and ordinary fixed-time convergence AF. Crucially, such

a predefined-time AF design facilitates better alignment with the time-delay term $\gamma_i(t)$ by allowing for the direct adjustment of the convergence time upper bound. Thus, for the design of AFs for predefined-time convergence, the key is to set the predefined-time parameters according to the time upper bound derived without gain coefficients after confirming the core function form.

By combining Eqs. (5) and (6), the NDRNN-CP proposed in this study can be derived.

$$u_i(t) = -\frac{g_i(t) + \gamma_i(t) - \sum_{j \in \mathcal{N}_i} u_j(t) - u_L(t)\delta(i)}{|\mathcal{N}_i| + \delta(i)}, \quad (8)$$

where $|\mathcal{N}_i|$ denotes the number of neighboring nodes of follower i , and $g_i(t) = \alpha_1 \phi(e_{f_l}(t)) + \alpha_2 \phi(e_{l_l}(t))$.

Remark 1. For a clearer demonstration of the advantages of NDRNN-CP (8), the mathematical expressions of both ODRNN-CP and IDRNN-CP are also presented in this paper for comparison.

- ODRNN-CP (designed with reference to Luo et al. (2023)):

$$u_i(t) = -\frac{g_i(t) - \sum_{j \in \mathcal{N}_i} u_j(t) - u_L(t)\delta(i)}{|\mathcal{N}_i| + \delta(i)}. \quad (9)$$

- IDRNN-CP (designed with reference to Jin et al. (2015), Liao et al. (2024)):

$$u_i(t) = -\frac{g_i(t) - \sum_{j \in \mathcal{N}_i} u_j(t) - u_L(t)\delta(i)}{|\mathcal{N}_i| + \delta(i)} - \frac{\mu_1 \int_0^t e_{f_l}(\tau) d\tau + \mu_2 \int_0^t e_{l_l}(\tau) d\tau}{|\mathcal{N}_i| + \delta(i)}, \quad (10)$$

where μ_1 and μ_2 are positive hyperparameters.

Regarding the parameter selection for the proposed NDRNN-CP, the trade-off between convergence rate and robustness is managed through a coordinated adjustment of the gains α_1, α_2 , the shaping exponent τ , the delay learning rate α_3 , and the scaling factor k . Initially, α_1 and α_2 set the fundamental response speed, while the introduction of τ refines this process by enabling a dual-regime convergence strategy. In this framework, the term with power $1 + \tau$ dominates during the reaching phase to accelerate the system when errors are large, whereas the term with power $1 - \tau$ provides an infinite-gain effect near the equilibrium to ensure strict finite-time convergence. Building upon this, α_3 and k are tuned to maintain stability against time-varying delays and uncertainties. Since higher gains generally improve tracking but may amplify high-frequency interference, a heuristic criterion is recommended where α_1 and α_2 are lowered slightly in high-noise environments, accompanied by an increased α_3 to prioritize noise rejection and delay adaptation. Conversely, in low-noise scenarios, increasing α_1 and α_2 is feasible to fully leverage the aggressive drive of the nonlinear power terms, thereby accelerating convergence with minimal risk of amplifying measurement noise.

4. Theoretical analysis

This section demonstrates the capability of NDRNN-CP (8) through a series of theoretical results. Initially, we investigate its convergence behavior in the absence of noise, which is detailed in Theorem 1. Subsequently, Theorem 2 focuses on the method's robustness when facing periodic disturbances. Finally, the predefined-time convergence of the NDRNN-CP (8) is discussed in Theorem 3.

Definition 1. (Xiao et al. (2024)) The Gamma function $\Gamma(a)$ is defined by the integral

$$\Gamma(a) = \int_0^{+\infty} t^{a-1} \exp(-t) dt,$$

where $a > 0$.

Definition 2. (Xiao et al. (2024)) The Beta function $B(a, b)$, with $a, b > 0$, is defined as

$$B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt.$$

Then, the relationship between the Gamma and Beta functions is given by:

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

4.1. Stability analysis

Consider a network with N agents whose communication structure is represented by a graph $G = (V, E, A)$. The vertex set $V = \{1, 2, \dots, N\}$ identifies the individual agents in the network, where each vertex corresponds to a single agent. The edge set $E \subseteq V \times V$ characterizes the communication topology: an edge $(i, j) \in E$ indicates that agents i and j can exchange information directly. The connectivity between agents is quantified by the adjacency matrix $A = [a_{ij}]_{N \times N}$, where $a_{ij} = 1$ if there exists a communication link from agent i to agent j , and $a_{ij} = 0$ otherwise. If the communication is bidirectional, the graph is undirected, meaning that the existence of an edge (i, j) implies the existence of (j, i) , and thus $a_{ij} = a_{ji}$.

The Laplacian matrix of the graph is given by $L = D - A$, where D is the degree matrix with diagonal entries $d_{ii} = \sum_{j=1}^N a_{ij}$. To incorporate the influence of a leader in leader-follower consensus scenarios, a diagonal matrix $B = \text{diag}(b_1, \dots, b_N)$ is introduced to represent leader accessibility: $b_i = 1$ if agent i can receive the leader's information, and $b_i = 0$ otherwise. The augmented matrix H is then defined as $H = L + B$. For an undirected and connected graph G , the matrix H is guaranteed to be positive definite provided that each $b_i \geq 0$ and at least one b_i is strictly greater than zero. This property ensures that the leader's information is propagated throughout the network and plays a crucial role in guaranteeing convergence in distributed control protocols.

Theorem 1. The NDRNN-CP (8) possesses global stability.

Proof. Consider the leader-follower part and $\gamma_i(t) = \gamma_i(t - T) + \alpha_3 \phi(e_i(t))$, we have

$$\dot{e}_l(t) = -\alpha_2 \phi(e_l(t)) - \gamma(t), \quad (11)$$

where $e_l(t) = [e_{l_1}(t), e_{l_2}(t), \dots, e_{l_N}(t)]^T$, $\gamma(t) = [\gamma_1(t), \gamma_2(t), \dots, \gamma_N(t)]^T$. It is worth noting that, unlike Eq. (4), in the present context we define $e_l(t) = x_l(t) - x_L(t)$. Then, a candidate Lyapunov function is defined as

$$V(t) = \frac{1}{2} e_l^T(t) H e_l(t) \geq 0, \quad (12)$$

and the derivative of $V(t)$ is obtained as follows:

$$\dot{V}(t) = e_l^T(t) H (-\alpha_2 \phi(e_l(t)) - \gamma(t)).$$

To further analyze $\dot{V}(t)$, the iterative property of $\gamma(t - T)$ introduced in Luo et al. (2021) is employed:

$$\begin{aligned} \dot{V}(t) &\leq e_l^T(t) H (-\alpha_2 \phi(e_l(t)) - \gamma(t - kT) - k\alpha_3 \phi(e_l(t))) \\ &\leq e_l^T(t) H (-\alpha_2 \phi(e_l(t)) - \mathbf{0}_N - k\alpha_3 \phi(e_l(t))) \\ &\leq -\lambda_{\min}(H)(\alpha_2 + k\alpha_3) e_l^T(t) \phi(e_l(t)), \end{aligned}$$

where $\lambda_{\min}(H)$ is the smallest eigenvalue of the matrix H . We know that $\phi(\cdot)$ is differentiable and strictly increasing near the origin. Then there exists a minimal slope $m_\phi > 0$ such that $\phi(x) \geq m_\phi x$. Therefore,

$$\dot{V}(t) \leq -m_\phi \lambda_{\min}(H)(\alpha_2 + k\alpha_3) e_l^T(t) e_l(t) \leq 0.$$

Since $V(t)$ is positive definite and $\dot{V}(t)$ is negative definite, the NDRNN-CP (8) is globally stable. $\square$

Remark 2. From the definition of $e_l(t)$, the term $(H e_l(t))_i$ is given by $\sum_{j \in \mathcal{N}_i} (x_i(t) - x_j(t)) + \delta(i) (x_i(t) - x_L(t))$. Here, the matrix H , defined as $L + B$, represents the augmented Laplacian. Provided that the graph is

connected and at least one node accesses the leader such that B is non-zero, H is positive definite. This mathematical property ensures that local neighbor-to-neighbor error minimization propagates through the network and forces the global leader-follower error vector $e_l(t)$ to zero. Although $e_l(t)$ involves all agents, only relevant components ultimately affect the dynamics. As $e_l(t)$ converges to $\mathbf{0}_N$, $H e_l(t)$ similarly converges to $\mathbf{0}_N$. In other words, the global error is implicitly accounted for via the positive definiteness of H even though the analysis focuses on the leader-follower error. Hence, $\gamma(t)$ can be independently added to $\dot{e}_l(t)$, and the reasoning in Theorem 2 and Theorem 3 follows accordingly.

4.2. Robustness analysis

In practice, external disturbances and measurement noise are inevitable. Hence, it is crucial to assess the robustness of NDRNN-CP (8) under bounded perturbations. The following analysis shows that its error dynamics converge to the origin despite such disturbances.

Theorem 2. Under the influence of the bounded noise vector $d(t) = [d_1(t), d_2(t), \dots, d_N(t)]^T$, the error vector of the NDRNN-CP (8) converges to $\mathbf{0}_N$ as time tends to infinity.

Proof. First, by adding noise $d(t)$ in Eq. (11), we have

$$\dot{e}_l(t) = -\alpha_2 \phi(e_l(t)) - \gamma(t) + d(t). \quad (13)$$

Then, a new candidate Lyapunov function can be defined as

$$L(t) = \frac{1}{2} e_l^T(t) H e_l(t) \geq 0. \quad (14)$$

Further, the derivative of $L(t)$ can be obtained as follows (Luo et al., 2021)

$$\begin{aligned} \dot{L}(t) &\leq e_l^T(t) H (-\alpha_2 \phi(e_l(t)) - \gamma(t - kT) \\ &\quad - k\alpha_3 \phi(e_l(t)) + d(t)) \\ &\leq -m_\phi \lambda_{\min}(H)(\alpha_2 + k\alpha_3) e_l^T(t) e_l(t) \\ &\quad + |e_l^T(t) H d(t)|. \end{aligned}$$

By further applying the Cauchy-Schwarz inequality (Wang et al., 2025c) and Young's inequality (Shakeel et al., 2025), the following inequality follows:

$$\begin{aligned} \dot{L}(t) &\leq -m_\phi \lambda_{\min}(H)(\alpha_2 + k\alpha_3) \|e_l(t)\|_2^2 \\ &\quad + \lambda_{\max}(H) \|e_l(t)\|_2 \|d(t)\|_2 \\ &\leq -m_\phi \lambda_{\min}(H)(\alpha_2 + k\alpha_3) \|e_l(t)\|_2^2 \\ &\quad + \frac{\lambda_{\max}(H)}{2\epsilon} \|d(t)\|_2^2 + \frac{\lambda_{\max}(H)\epsilon}{2} \|e_l(t)\|_2^2, \end{aligned} \quad (15)$$

where $\epsilon > 0$. Let $k_{e_l} = m_\phi \lambda_{\min}(H)(\alpha_2 + k\alpha_3)$, $M = \lambda_{\max}(H)$. Two cases are considered for $\dot{L}(t)$ in the presence of noise $d(t)$.

1. If $\|d(t)\|_2 \equiv 0$, it follows from Theorem 1 that $\dot{L} \leq 0$.
2. If $\|d(t)\|_2 \leq \zeta_d$ with $\zeta_d > 0$, then Eq. (15) is rewritten as

$$\dot{L}(t) \leq -\left(k_{e_l} - \frac{M\epsilon}{2}\right) \|e_l(t)\|_2^2 + \frac{M}{2\epsilon} \zeta_d^2,$$

where ϵ must satisfy $0 < \epsilon < \frac{2k_{e_l}}{M}$, which ensures that $\varpi = (k_{e_l} - \frac{M\epsilon}{2}) > 0$. Moreover, if $\|e_l(t)\|_2^2 \geq \frac{M}{2\epsilon\varpi} \zeta_d^2$ then

$$\dot{L}(t) \leq -\varpi \|e_l(t)\|_2^2 + \frac{M}{2\epsilon} \zeta_d^2 \leq 0. \quad (16)$$

It is worth noting that if $\dot{L}(t) > 0$, then $L(t)$ will increase, and $\dot{L}(t)$ will decrease to zero. Therefore, the upper bound of $e_l(t)$ can be derived as follows:

$$\limsup_{t \rightarrow \infty} \|e_l(t)\|_2 \leq \zeta_d \sqrt{\frac{M}{2\epsilon\varpi}}.$$

$\square$

In addition to asymptotic and robust convergence, it is often desirable to guarantee that the system reaches stability within a fixed and user-specified time, independent of initial conditions. To this end, we analyze the predefined-time convergence property of NDRNN-CP (8).

Theorem 3. *Given that T_c is the convergence time and T_p is a predefined-time parameter, the NDRNN-CP (8) can achieve predefined-time convergence from any initial value $e_i(0)$.*

Proof. According to Theorem 1, the i -th component $e_{i_i}(t)$ of the error vector $e_i(t)$ will converge to zero with the bounded noise $d_i(t)$, and we have

$$\begin{aligned}\dot{e}_{i_i}(t) &= -\alpha_2 \phi(e_{i_i}(t)) - \gamma_i(t) + d_i(t) \\ &= -\frac{\pi \alpha_2}{k \sqrt{\beta_1 \beta_2}} \left(\beta_1 |e_{i_i}(t)|^{1-\tau} + \beta_2 |e_{i_i}(t)|^{1+\tau} \right) - \gamma_i(t) + d_i(t).\end{aligned}$$

If $|e_{i_i}(t)|$ converges to zero after time T_c and the condition $(\gamma_i(t) - d_i(t)) \text{sign}(e_{i_i}(t)) \geq 0$ is satisfied, which ensures that the external disturbance signal d_i is always within the suppression range of γ_i , so that the noise disturbance can be consistently suppressed whether $e_{i_i}(t)$ is in the negative or positive bound. The derivative of $|e_{i_i}(t)|$ is given by:

$$\begin{aligned}\frac{d|e_{i_i}(t)|}{dt} &= \dot{e}_{i_i}(t) \text{sign}(e_{i_i}(t)) \\ &= \left(-\alpha_2 \phi(e_{i_i}(t)) - \gamma_i(t) + d_i(t) \right) \text{sign}(e_{i_i}(t)) \\ &\leq -\alpha_2 \phi(e_{i_i}(t)) \text{sign}(e_{i_i}(t)) \\ &\leq -\alpha_2 \phi(|e_{i_i}(t)|).\end{aligned}$$

Then,

$$\begin{aligned}dt &\leq -\frac{1}{\alpha_2 \phi(|e_{i_i}(t)|)} d|e_{i_i}(t)| \\ &= -\frac{k \sqrt{\beta_1 \beta_2} d|e_{i_i}(t)|}{\pi \alpha_2 \left(\beta_1 |e_{i_i}(t)|^{1-\tau} + \beta_2 |e_{i_i}(t)|^{1+\tau} \right)}.\end{aligned}\quad (17)$$

It is obvious that T_c can be described as

$$T_c = \int_0^{T_c} 1 dt. \quad (18)$$

According to the integral inequality and by combining Eqs. (17) and (18), it follows that:

$$\begin{aligned}T_c &\leq \int_{|e_{i_i}(0)|}^{|e_{i_i}(T_c)|} -\frac{k \sqrt{\beta_1 \beta_2} d|e_{i_i}(t)|}{\pi \alpha_2 \left(\beta_1 |e_{i_i}(t)|^{1-\tau} + \beta_2 |e_{i_i}(t)|^{1+\tau} \right)} \\ &\leq \frac{k \sqrt{\beta_1 \beta_2}}{\pi \alpha_2} \int_0^{+\infty} \frac{d|e_{i_i}(t)|}{\left(\beta_1 |e_{i_i}(t)|^{1-\tau} + \beta_2 |e_{i_i}(t)|^{1+\tau} \right)} \\ &= \frac{k \sqrt{\beta_1 \beta_2}}{\pi \alpha_2} \int_0^{+\infty} \frac{dc}{\left(\beta_1 c^{1-\tau} + \beta_2 c^{1+\tau} \right)},\end{aligned}\quad (19)$$

where $c = |e_{i_i}(t)| \neq 0$.

By letting $v = \frac{\beta_2}{\beta_1} c^{2\tau}$, we obtain the following derivation:

$$\begin{aligned}I &= \int_0^{+\infty} \frac{dc}{\beta_1 c^{1-\tau} + \beta_2 c^{1+\tau}} \\ &= \frac{1}{\beta_1} \int_0^{+\infty} \frac{c^{\tau-1}}{1 + (\beta_2/\beta_1) c^{2\tau}} dc \\ &= \frac{1}{2\tau \beta_1} \left(\frac{\beta_1}{\beta_2} \right)^{1/(2\tau)} \int_0^{+\infty} \frac{c^{\tau-1} v^{1/(2\tau)-1}}{1+v} dv \\ &= \frac{1}{2\tau \beta_1} \left(\frac{\beta_1}{\beta_2} \right)^{1/(2\tau)} \int_0^{+\infty} \frac{\left(\frac{\beta_1}{\beta_2} v \right)^{(\tau-1)/(2\tau)} v^{1/(2\tau)-1}}{1+v} dv\end{aligned}$$

$$= \frac{1}{2\tau \sqrt{\beta_1 \beta_2}} \int_0^{+\infty} \frac{v^{-1/2}}{1+v} dv. \quad (20)$$

Let $w = \frac{v}{1+v}$. From this, $v = \frac{w}{1-w}$ and $dv = \frac{dw}{(1-w)^2}$. Consequently, it can be deduced that

$$\begin{aligned}I &= \frac{1}{2\tau \sqrt{\beta_1 \beta_2}} \int_0^1 \frac{(1-w) \left(\frac{w}{1-w} \right)^{-1/2}}{(1-w)^2} dw \\ &= \frac{1}{2\tau \sqrt{\beta_1 \beta_2}} \int_0^1 w^{-1/2} (1-w)^{-1/2} dw \\ &= \frac{1}{2\tau \sqrt{\beta_1 \beta_2}} \mathcal{B}\left(\frac{1}{2}, \frac{1}{2}\right) \\ &= \frac{1}{2\tau \sqrt{\beta_1 \beta_2}} \frac{\Gamma\left(\frac{1}{2}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2} + \frac{1}{2}\right)} \\ &= \frac{\pi}{2\tau \sqrt{\beta_1 \beta_2}}.\end{aligned}\quad (21)$$

Furthermore, from Definitions 1 and 2, it follows that

$$\begin{aligned}T_c &\leq \frac{k \sqrt{\beta_1 \beta_2}}{\pi \alpha_2} I \\ &= \frac{k}{2\tau \alpha_2}.\end{aligned}\quad (22)$$

Finally, if $2\tau = 1$, the predefined-time upper bound T_p is obtained as follows:

$$T_c \leq \frac{k}{\alpha_2} = T_p.$$

That is, the i -th component $e_{i_i}(t)$ of the error vector $e_i(t)$ will converge to zero within the predefined-time T_p under bounded noise $d_i(t)$. It follows that $e_i(t)$ also converges to $\mathbf{0}_N$ within the predefined-time T_p . The proof is complete. $\square$

5. Consensus control experimental evaluation

In this section, we present a comprehensive evaluation of the proposed NDRNN-CP (8) and its extension, NDRNN-S (26), through both coordination and sensing tasks. For the 2-D multi-agent consensus experiments, three settings are considered: (i) a baseline without noise, (ii) noise-affected cases using the disturbance signals in Fig. 1 to evaluate robustness, and (iii) an expanded scenario with the number of agents increased from 10 to 32 to examine scalability. In all cases, NDRNN-CP (8) is compared with ODRNN-CP (9) and IDRNN-CP (10), with the expanded experiments additionally assessing the impact of different activation functions (AFs), on convergence speed and steady-state accuracy. To verify applicability in sensing, a distributed acoustic source localization (DASL, also known as sound source localization) (Gao et al., 2023; Krause et al., 2024) experiment is conducted in a 3-D environment based on the TDOA (Dang et al., 2022) principle, where NDRNN-S (26) is compared with ODRNN-S (28) and IDRNN-S (29) under both noise-free and noisy conditions, also including different AFs in the analysis. The results demonstrate the proposed methods' strong convergence performance, robustness to disturbances, and adaptability to both large-scale multi-agent coordination and distributed sensing tasks.

In this experiment, there are 10 agents in total, including 1 leader and 9 followers. The communication topology is illustrated in Fig. 2, where the leader has direct links only with followers 1, 2, 5, 8, and 9. The leader follows a butterfly-shaped path in the 2-D plane, which can be described mathematically as follows:

$$\mathbf{x}_L(t) = \begin{bmatrix} 50 \sin(t) \left(\exp(\cos(t)) - 2 \cos(4t) - \sin^5\left(\frac{t}{12}\right) \right) \\ 50 \cos(t) \left(\exp(\cos(t)) - 2 \cos(4t) - \sin^5\left(\frac{t}{12}\right) \right) \end{bmatrix}.$$

The experimental results are shown in Figs. 3–8. The hyperparameters are chosen as $\alpha_1 = \alpha_2 = 10$, $\alpha_3 = \mu_1 = \mu_2 = 50$, $\beta_1 = \beta_2 = 1$, $k = 8$ and

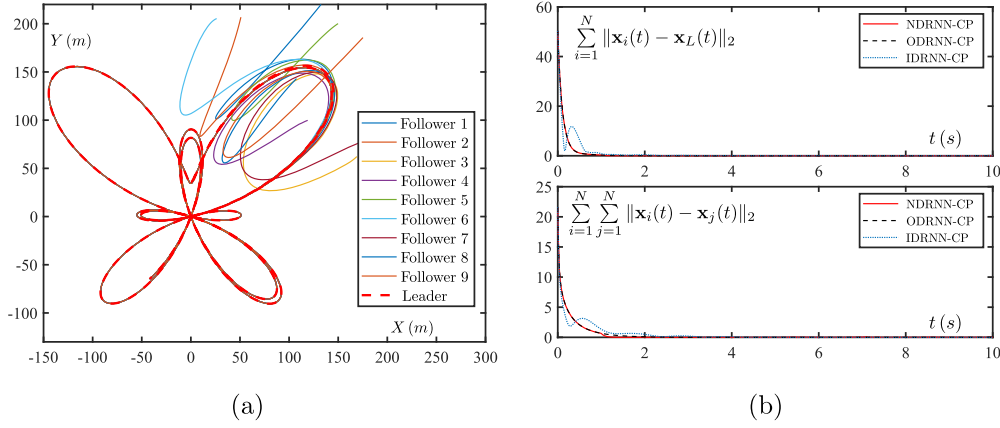


Fig. 3. Results of the multi-agent consensus control experiment without noise signal interference. (a) Motion trajectories of agents in the 2-D space using NDRNN-CP (8). (b) Steady-state errors for different control protocols.

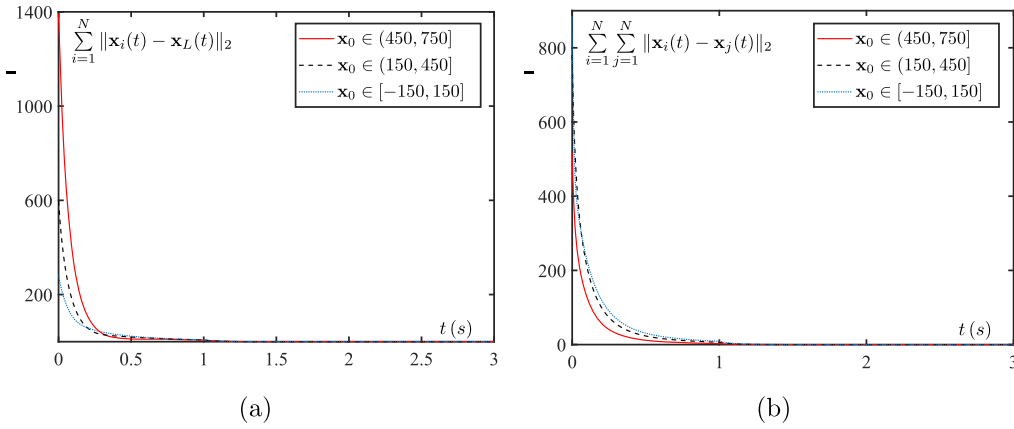


Fig. 4. Steady-state errors with different initial states $\mathbf{x}_0$ for the NDRNN-CP (8). (a) Steady-state error $\sum_{i=1}^N \|\mathbf{x}_i(t) - \mathbf{x}_L(t)\|_2$. (b) Steady-state error $\sum_{i=1}^N \sum_{j=1}^N \|\mathbf{x}_i(t) - \mathbf{x}_j(t)\|_2$.

$\tau = 0.5$. However, in the distributed acoustic source localization experiment, k is adjusted to 1 to better adapt to the 3-D task scenario. Notably, the parameter k functions as a scaling factor that inversely modulates the upper bound of the convergence time. This relationship implies that a larger k leads to a more relaxed convergence process, whereas a smaller k enforces a stricter and faster convergence rate. Specifically, $k = 8$ is selected for the consensus task to achieve a balance between convergence speed and overshoot suppression. In contrast, for the DASL task, k is adjusted to 1 to prevent numerical instability and accommodate the high-precision requirements of the 3-D sensing domain.

5.1. Noise-free experiment

This subsection examines the performance of each control protocol for the multi-agent system in the absence of noise. The results are shown in Figs. 3 and 4. In Fig. 3(a), the motion trajectories under noise-free conditions are displayed. Starting from random initial positions, the nine followers rapidly adjust their states to match that of the leader as time advances, demonstrating the capability of NDRNN-CP (8) to achieve consensus. Fig. 3(b) further indicates that all three control protocols perform well, as the total state error across all agents steadily decreases and eventually reaches zero.

As shown in Figs. 4(a)-(b), even when the agents begin with large differences in their initial states, the steady-state error still converges to zero in roughly the same time. This confirms that NDRNN-CP (8) can drive the multi-agent system to consensus within a predefined time, ensuring fast state alignment despite significant initial disparities.

5.2. Noise-affected experiment

In this section, the performance of the three control protocols is evaluated under different types of noise disturbances. The experimental outcomes are presented in Figs. 5 and 6, with the noise amplitude set to 50 and the noise waveforms shown in Fig. 1.

Figs. 5(a)-(c) show that the NDRNN-CP (8) effectively suppresses periodic disturbances, including the sine, square, and sawtooth waves, enabling the total state error across the multi-agent system to converge to zero. Under the same conditions, the ODRNN-CP (9) and the IDRNN-CP (10) fail to achieve complete convergence. In Fig. 5(d), for the constant noise, both the IDRNN-CP (10) and the NDRNN-CP (8) stabilize the system, with the NDRNN-CP (8) converging slightly faster. Figs. 5(e) and 5(f) (zoomed view) show that while the zero-mean random noise prevents all three protocols from fully eliminating the steady-state error, the NDRNN-CP (8) attains much smaller residual errors, indicating stronger resistance to random disturbances than the ODRNN-CP (9) and the IDRNN-CP (10).

Fig. 6 presents the comparison of the ODRNN-CP (9), the IDRNN-CP (10), and the NDRNN-CP (8) under the composite periodic noise. As seen in Figs. 6(a)-(c), the ODRNN-CP (9) and the IDRNN-CP (10) exhibit noticeable deviation from the leader's path, while the NDRNN-CP (8) enables the agents to closely track the butterfly-shaped trajectory with minimal error, showing superior accuracy under complex noise conditions. Fig. 6(d) further illustrates the robustness of the NDRNN-CP (8), maintaining low steady-state error despite the presence of noise.

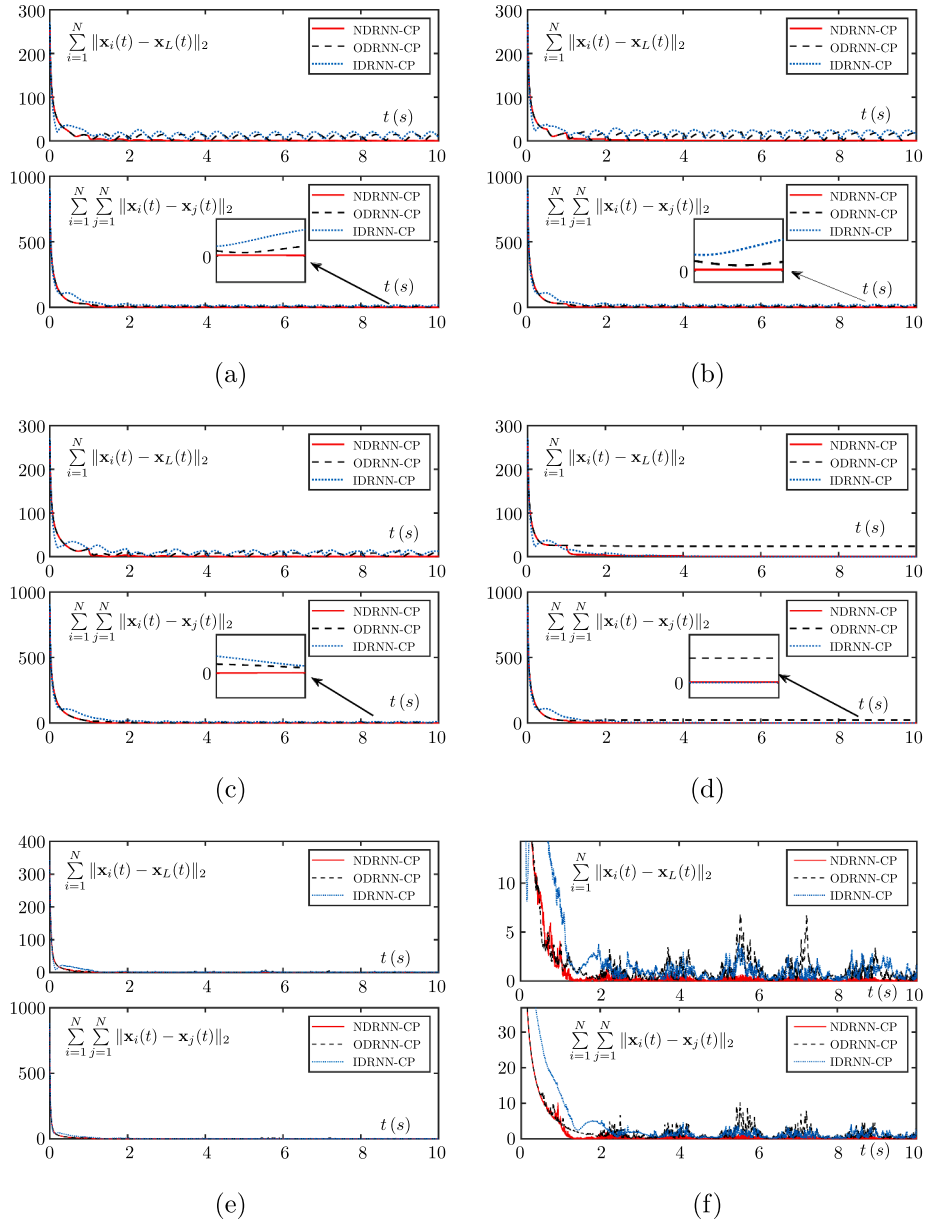


Fig. 5. Steady-state errors for different control protocols under various noise signal disturbances. (a) Sine wave. (b) Square wave. (c) Sawtooth wave. (d) Constant. (e) Zero-mean random. (f) Magnified view of (e).

Overall, the results confirm that the NDRNN-CP (8) can achieve consensus within a predefined time and effectively counter both periodic and random noise disturbances, ensuring system stability and accurate state alignment.

5.3. Expanded experiment

This subsection examines the scalability and effectiveness of the control protocols by expanding the number of agents from 10 to 32, where the 32 agents (1 leader and 31 followers) communicate randomly while ensuring the topology remains connected. where the 32 agents (1 leader and 31 followers) communicate randomly based on the Erdős-Rényi graph model with a connectivity probability of $p = .4$ while ensuring the topology remains connected. The results in Figs. 7 and 8 indicate that the proposed control approaches remain effective under both noise-free and composite noise conditions, demonstrating their robustness and ability to achieve predefined-time convergence in larger-scale multi-agent networks.

For the improvement based on Eq. (7), which specifies the proposed AF, a comparison is made with the widely used linear and Sign-Bi-Power (SBP) AFs (Liao et al., 2025). As illustrated in Figs. 7(a)-(b), the proposed AF yields significantly faster convergence and lower steady-state error, thereby verifying its effectiveness.

Figs. 8(a)-(b) further show that, even with noise present, all agents guided by the NDRNN-CP (8) converge to a common state over time, highlighting the capability of method to handle large-scale and complex multi-agent scenarios.

Remark 3. The implementation of the proposed NDRNN-CP fundamentally relies on the distributed nature of the Laplacian matrix. This distributed characteristic enables the NDRNN-CP/NDRNN-S to accommodate dynamic topological structures while ensuring global stability. While standard consensus theories typically suggest that error states will eventually converge as long as the communication graph remains connected on average over time (i.e., under dynamic switching topologies), the NDRNN-CP leverages its intrinsic real-time-varying solving

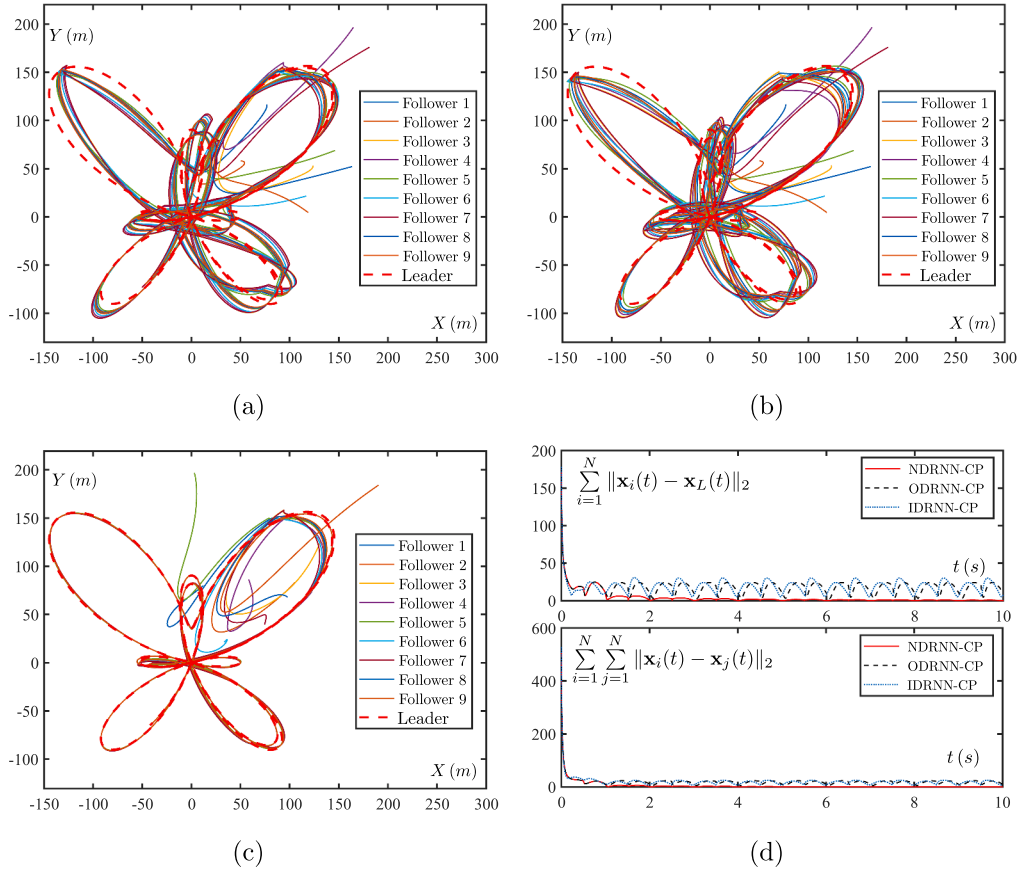


Fig. 6. Results of the multi-agent consensus control experiment under composite periodic noise signal interference. (a) Motion trajectories of agents with ODRNN-CP (9). (b) Motion trajectories of agents with IDRNN-CP (10). (c) Motion trajectories of agents with NDRNN-CP (8). (d) Steady-state errors for different control protocols.

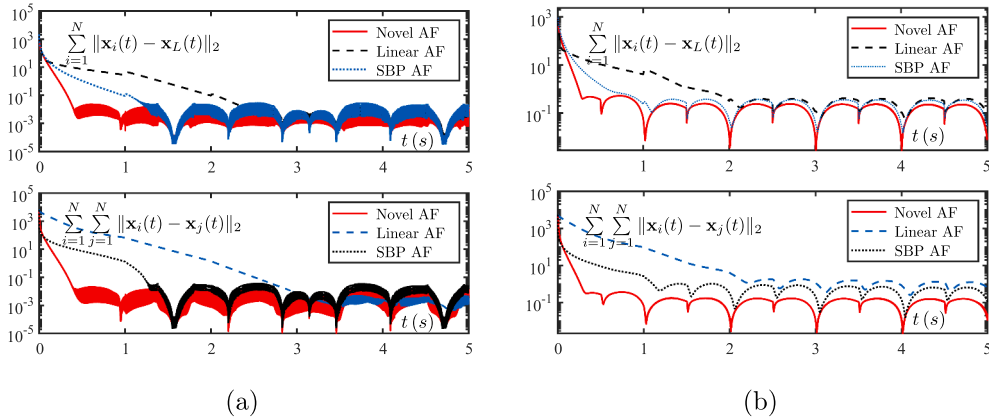


Fig. 7. Steady-state errors in the expanded experiment of 32 agents with different AFs. (a) Different AFs in NDRNN-CP (8) on noise-free signal. (b) Different AFs in NDRNN-CP (8) on composite noise signal.

characteristic to offer a more rigorous guarantee. Specifically, even in the event of dynamic topology changes or temporary communication interruptions, provided that the network remains jointly connected within a practically permissible timeframe, the protocol ensures not only the final error convergence but also explicitly maintains its predefined-time convergence property. This capability allows the system to exhibit superior robustness compared to asymptotic convergence methods in dynamic environments. Complementing these theoretical robustness guarantees, the proposed framework also ensures computational efficiency.

While the incorporated delay mechanism and the new activation function introduce a minor additional computational complexity compared to the baseline DRNN, this increase is marginal. The delay mechanism requires only a simple buffer with $O(1)$ additional memory storage, and the activation function is computed element-wise with a constant calculation time. Furthermore, through its distributed design, since each agent i computes Eq. (8) relying solely on information from its local neighbors $|\mathcal{N}_i|$, the computational cost per agent scales as $O(|\mathcal{N}_i|)$ rather than the total network size $O(N)$. Consequently, these constant-level

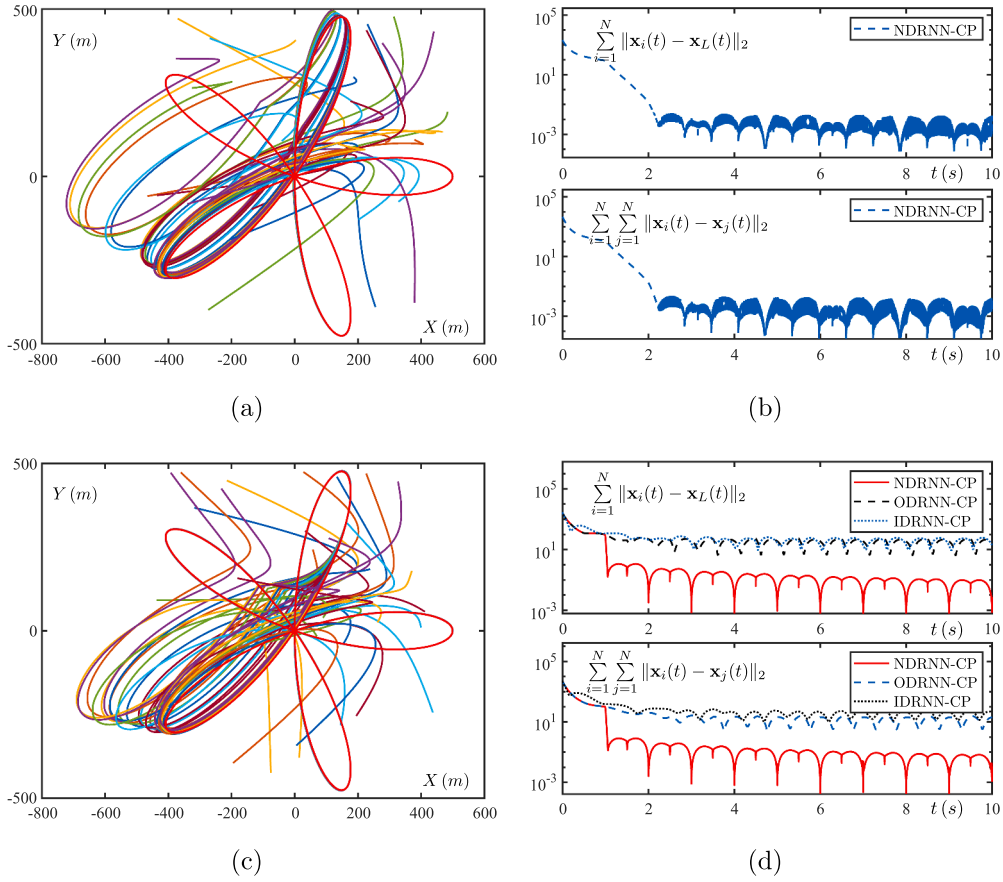


Fig. 8. Trajectories and steady-state errors of 32 agents in the expanded experiment. (a) Motion trajectories under noise-free signal. (b) Steady-state errors under noise-free signal. (c) Motion trajectories under composite noise signal. (d) Steady-state errors under composite noise signal.

additions do not alter the overall order of complexity, preventing complexity explosion as the number of agents increases and thoroughly ensuring the scalability of the system.

6. NDRNN for distributed acoustic source localization

In this section, based on the TDOA principle, we evaluate the performance of two existing solvers as well as the proposed NDRNN-S (26) in DASL within a 3-D environment. Notably, in DASL, each receiving node exchanges data with its neighboring nodes instead of only with a fixed unit. In addition, we analyze the localization performance of the NDRNN-S (26) under the influence of the proposed AF and two other AFs. The experiments consider both noise-free and noise-affected scenarios to comprehensively assess the robustness and adaptability of the NDRNN-S (26).

In the DASL task, the acoustic source acts as the signal emitter, while microphones serve as signal receivers. By computing the time differences between the emitted signal and the signals received by each microphone, the localization problem can be transformed into solving a time-varying linear system (Jin et al., 2020). We define the positions of the m microphones and the acoustic source in a 3-D space as follows:

$$M = \begin{bmatrix} x_1 & x_2 & \cdots & x_m \\ y_1 & y_2 & \cdots & y_m \\ z_1 & z_2 & \cdots & z_m \end{bmatrix} \in \mathbb{R}^{3 \times m}, \quad S(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix} \in \mathbb{R}^3.$$

Each column of M represents the position of a microphone, while $S(t)$ is the time-varying position of the acoustic source. The microphones are fixed in space and are uniformly randomly distributed around the time-varying position of the acoustic source.

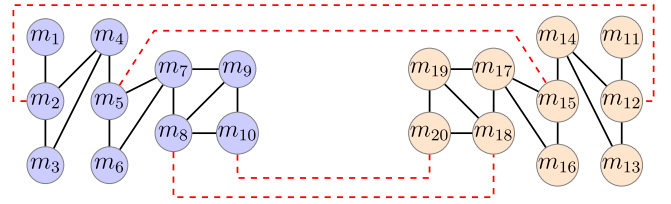


Fig. 9. The microphone communication topology.

Let the undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ represent the microphone communication topology, where $|\mathcal{V}| = 20$. In this topology, each node communicates only with its directly connected neighbors specified in $\mathcal{E}$.

As shown in Fig. 9, the communication topology among the 20 microphones is illustrated, where each node represents a microphone and edges denote direct communication links. Blue nodes correspond to microphones $m_1 - m_{10}$, and orange nodes correspond to microphones $m_{11} - m_{20}$. Dashed red lines indicate cross-layer connections between the two mirrored groups, using orthogonal paths to avoid overlapping. It can be observed that the edge set $\mathcal{E}$ contains no duplicate edges. This design reduces communication overhead while preserving local connectivity, and clearly defines the communication links for efficient information exchange among neighboring microphones in DASL.

To simulate the trajectory of the moving acoustic source, this paper adopts the classical Lissajous curve, which is mathematically expressed in three dimensions as follows:

$$x(t) = 5 \sin(3t + \frac{\pi}{2}), \quad y(t) = 5 \sin(4t), \quad z(t) = 5 \sin(5t).$$

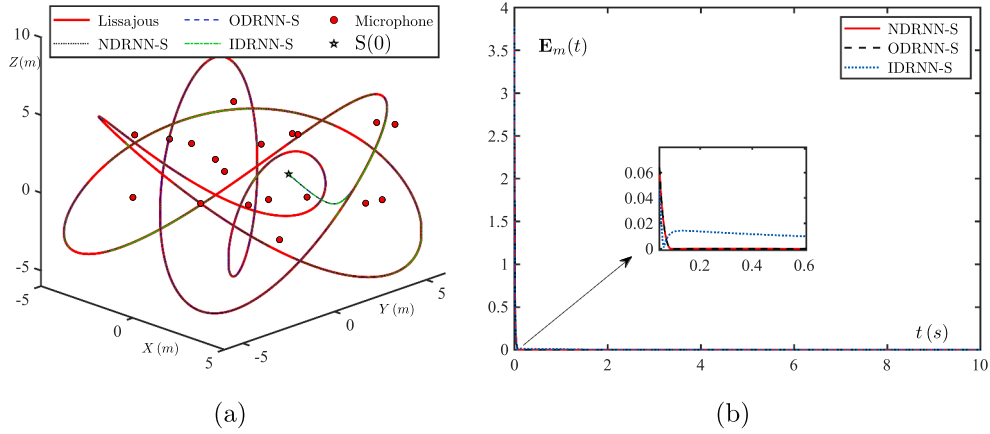


Fig. 10. Results of the DASL experiment without noise signal interference. (a) Trajectories of acoustic source positions localized by different solvers. (b) Steady-state errors of different solvers.

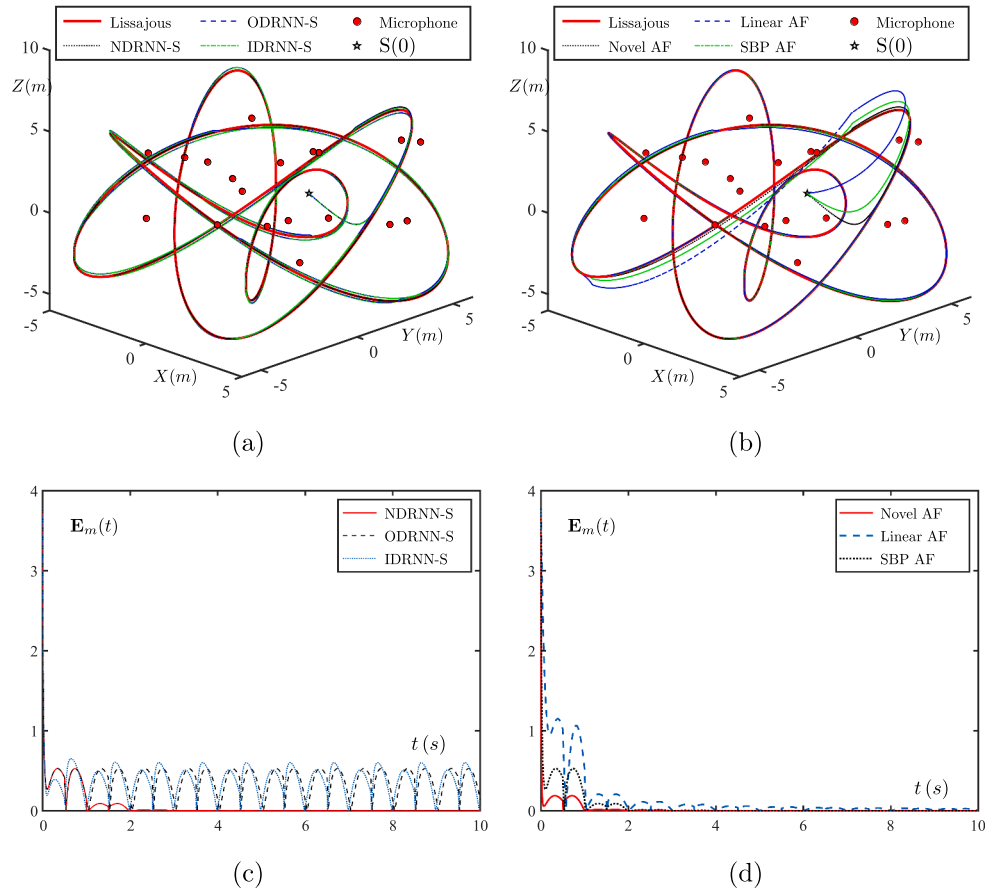


Fig. 11. Results of the DASL experiment under composite periodic noise signal interference. (a) Trajectories of acoustic source positions localized by different solvers. (b) Trajectories of acoustic source positions localized by NDRNN-S (26) with different AFs. (c) Steady-state errors of (a). (d) Steady-state errors of (b).

We design the following equations:

$$d_i(t) = \|S(t) - M_i\|_2, \Delta T_{ij}(t) = T_i(t) - T_j(t),$$

where $i \in \{1, \dots, m\}$, $d_i(t)$ is the distance between the source and the i -th microphone, and $\Delta T_{ij}(t)$ is the TDOA between microphones i and j .

Then, according to the above, the TDOA can be expressed in terms of the corresponding distance difference as follows:

$$v\Delta T_{ij}(t) = d_i(t) - d_j(t), \quad (23)$$

where $v = 340.29$ m/s is the speed of sound in air at standard conditions. Squaring both sides of Eq. (23) and expanding the distances yields:

$$\begin{aligned} & (x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2 - v^2 \Delta T_{ij}(t)^2 \\ &= (x - x_j)^2 + (y - y_j)^2 + (z - z_j)^2 + 2v\Delta T_{ij}(t)d_j(t). \end{aligned}$$

Then it can be derived that:

$$\begin{aligned} & 2(x_j - x_i)x + 2(y_j - y_i)y + 2(z_j - z_i)z \\ &= x_j^2 + y_j^2 + z_j^2 - x_i^2 - y_i^2 - z_i^2 + 2v\Delta T_{ij}(t)d_j(t) + v^2 \Delta T_{ij}(t)^2. \end{aligned}$$

We define the coefficients as follows:

$$\begin{aligned} a_{ij,1} &= 2(x_j - x_i), \quad a_{ij,2} = 2(y_j - y_i), \quad a_{ij,3} = 2(z_j - z_i). \\ b_{ij}(t) &= x_j^2 + y_j^2 + z_j^2 - x_i^2 - y_i^2 - z_i^2 + 2v^2 \Delta T_{ij}(t) T_j(t) \\ &\quad + v^2 \Delta T_{ij}(t)^2. \end{aligned}$$

Finally, for each $j \in \mathcal{N}_i$ given node i and its neighbor set $\mathcal{N}_i$, a local linear system can be derived:

$$A_i \mathbf{S}(t) = b_i(t),$$

where $A_i = [a_{ij,1}, a_{ij,2}, a_{ij,3}] \in \mathbb{R}^{|\mathcal{N}_i| \times 3}$, and $b_i(t) \in \mathbb{R}^{|\mathcal{N}_i|}$.

Aggregating all local systems, we can obtain the global linear system:

$$H \mathbf{S}(t) = \mathbf{G}(t), \quad (24)$$

where H is composed of all A_i for each node i , and $\mathbf{G}(t)$ includes the corresponding $b_i(t)$. Notably, this transformation converts the non-convex intersection problem into a linear system by introducing auxiliary variables, effectively avoiding the need to directly solve a global non-convex optimization problem. Nevertheless, the task remains challenging due to the time-varying nature of the system, the presence of noise, and the constraints imposed by distributed information. Therefore, the proposed NDRNN-S is specifically designed to robustly solve these time-varying linear residual dynamics in a distributed manner with periodic disturbance compensation, rather than addressing the original non-convex formulation directly.

According to the design process of the NDRNN, the following error function can be defined:

$$\mathbf{E}_m(t) = H \mathbf{S}(t) - \mathbf{G}(t), \quad (25)$$

Then, we can derive a novel DRNN-based solver (NDRNN-S):

$$\dot{\mathbf{E}}_m(t) = -\alpha_1 \Phi(\mathbf{E}_m(t)) - \gamma_m(t), \quad (26)$$

where $\Phi(\cdot) : \mathbb{R}^{3 \times 1} \rightarrow \mathbb{R}^{3 \times 1}$, referred to as the AF, is a vector mapping composed of odd and monotonically increasing functions $\phi(\cdot)$.

The specific form of Eq. (26) is given by:

$$H \dot{\mathbf{S}}(t) = -\alpha_1 \phi(H \mathbf{S}(t) - \mathbf{G}(t)) + \dot{\mathbf{G}}(t) - \gamma_m(t), \quad (27)$$

where

$$\gamma_m(t) = \begin{cases} 0 & \text{if } t < T, \\ \gamma_m(t - T) + \alpha_3 \phi(\mathbf{E}_m(t)) & \text{otherwise,} \end{cases}$$

T is the noise period.

Based on the ODRNN-CP (9) and IDRNN-CP (10), the following two solvers are selected for comparison:

- Original DRNN-based Solver (ODRNN-S):

$$\dot{\mathbf{E}}_m(t) = -\alpha_1 \Phi(\mathbf{E}_m(t)). \quad (28)$$

- Integral-type DRNN-based Solver (IDRNN-S):

$$\dot{\mathbf{E}}_m(t) = -\alpha_1 \Phi(\mathbf{E}_m(t)) - \mu_1 \int_0^t \mathbf{E}_m(t) dt. \quad (29)$$

In the following experiments, 20 microphones are selected, and the initial position of the acoustic source is set as $\mathbf{S}(0) = [1.5, 1, 1]^T$. The composite periodic noise shown in Fig. 1 is introduced as signal interference. Figs. 10 and 11 demonstrate the effectiveness of the proposed NDRNN-S (26) in solving the acoustic source localization problem. Fig. 10a shows that all three solvers can accurately locate the acoustic source under noise-free conditions. Fig. 10b further compares the steady-state residual convergence of the three solvers, where IDRNN-S (29) performs slightly worse in the early stage.

Fig. 11a shows the estimated trajectories of the time-varying acoustic source under composite noise, where the NDRNN-S (26) achieves higher accuracy than the other two solvers. The robustness of NDRNN-S (26) against noise is further confirmed in Fig. 11c. To examine the impact of AFs, two common AFs are compared with the proposed one: Fig. 11b presents the localization results, while Fig. 11d demonstrates that the proposed AF yields superior steady-state residual convergence.

7. Conclusion

This paper proposes a NDRNN framework and its consensus control protocol (NDRNN-CP) for multi-agent systems. The design incorporates a time-delay component to adaptively capture noise variation patterns and embeds signal processing within the control framework, enabling active suppression of both periodic and random disturbances. A tailored AF was developed to guarantee predefined-time convergence, and rigorous analysis established global stability, robustness to bounded disturbances, and convergence within the prescribed time. Extensive simulations and experiments demonstrated that NDRNN-CP achieves rapid and reliable consensus even in large-scale networks with up to 32 agents, outperforming mainstream DRNN-based methods in both convergence speed and noise resistance. Furthermore, the NDRNN-based solver (NDRNN-S) for distributed acoustic source localization achieved superior accuracy and robustness in noisy and reverberant environments, confirming the broad applicability of the proposed framework to complex multi-agent coordination and distributed sensing tasks.

CRedit authorship contribution statement

Lei Jia: Funding acquisition, Methodology, Writing – original draft; **Weibao Xiong:** Formal analysis, Software, Writing – original draft; **Lin Xiao:** Validation; **Yiwei Li:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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